



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

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Lab Assignment 1: Daily Activities Recorder App

SECP3106 APPLICATION DEVELOPMENT (WBL)

SECTION 01

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1. Objective

The primary objective of this assignment is to apply the basic concepts of Flutter development learned in Lessons 1 through 3. This involves understanding the Flutter project structure, utilizing layout widgets, capturing text input, displaying dynamic lists, and implementing state management using StatefulWidget and the setState() method.

2. Brief description of the system

The Daily Activities Recorder is a simple Flutter application that allow users to enter activity information and display the recorded activities on the screen. It operates without a complex backend database, the system relies entirely on local state management to provide immediate UI updates. The system supports full CRUD (Create, Read, Update, Delete) operations, allowing the user add and view daily activities and edit existing activities and safely delete them. The system features a clean, modern interface designed for clarity and ease of use.

The app allow the user to:

1. Enter a daily activity
2. Enter the time or category of the activity
3. Save the activity
4. Display the list of recorded activities
5. Delete an activity from the list

3. Explanation of the interface

The user interface is designed with user-friendly, with a top input form and a scrollable activity list below.

A screenshot of a single text input field with the word "Breakfast" entered. The field has a light gray border and rounded corners.

- **Activity Input Field:** A TextField widget designed to capture the primary name of the task the user wishes to perform (e.g., "Study Flutter", "Breakfast").

A screenshot of a text input field with the word "Jogging" entered. The field has a light gray border and rounded corners.A screenshot of a text input field with the text "8:00 PM/Exercise" entered. The field has a light gray border and rounded corners.

- **Time/Category Input Field:** A secondary TextField that allows the user to provide additional context, such as a specific time (e.g., "8:00 PM") or a broad category (e.g., "Exercise").

Update Activity

Cancel

+ Add Activity

Clear

- Add Activity / Update Button:** An ElevatedButton that manages data saving. When adding a new activity, tapping this button triggers setState() to append the text field data into the application's list, immediately refreshing the screen. This button is dynamic; if the user is modifying an existing record, the text changes to "Update Activity" and saves the new values to the selected item.

Update Activity

Cancel

+ Add Activity

Clear

- Clear / Cancel Button:** An OutlinedButton used to empty the input fields. If the app is currently in edit mode, this button changes to "Cancel" and safely exits the editing process without altering the original data.

ACTIVITIES

Jogging

8:00 PM/Exercise

Breakfast

9.00AM/Meal

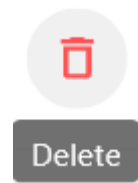
Lunch

1.00PM/Meal

- Activity List:** Displayed using a ListView.builder , this section renders all recorded activities as individual styled cards. It is wrapped in an Expanded widget to ensure it takes up the remaining screen space and scrolls smoothly.



- **Edit Button:** A black IconButton located on each activity card. Tapping it loads the selected activity's details back into the text fields, transitioning the app into "Edit Mode".



- **Delete Button:** A red IconButton used to remove an activity. To prevent accidental data loss, tapping this button triggers an AlertDialog asking the user to confirm their action before the item is permanently deleted from the list.

A screenshot of a mobile application titled "Daily Activities Recorder". The interface features a top blue header bar. Below it, there are two input fields: the first contains "Jogging" and the second contains "8:00 PM/Exercise". Below these fields are two buttons: a blue "Update Activity" button and a white "Cancel" button with a blue border. Underneath the buttons is a section titled "ACTIVITIES" which contains a list of three activity cards. Each card displays the activity name, time, and type, along with edit and delete icons. The first card is for "Jogging" at "8:00 PM/Exercise". The second is for "Breakfast" at "9:00AM/Meal". The third is for "Lunch" at "1:00PM/Meal".

4. Conclusion

This assignment successfully demonstrates the fundamental concepts of Flutter app development. I combine TextField inputs with a ListView and utilizing setState() to handle the data, which results in an app with a smooth experience for activity tracking. This assignment include advanced logic such as dynamic update system and the deletion confirmation dialog proves that the system is strong, user-friendly, and can effectively manage the complex state changes.